Dynamics Analysis of a manipulator-like spacecraft in the close range approach for rendezvous and docking/berthing by using the Lagrangian formulation with the Lagrange's multiplier approach to account for the reaction forces/torque in the manipulator joints.

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Abstract

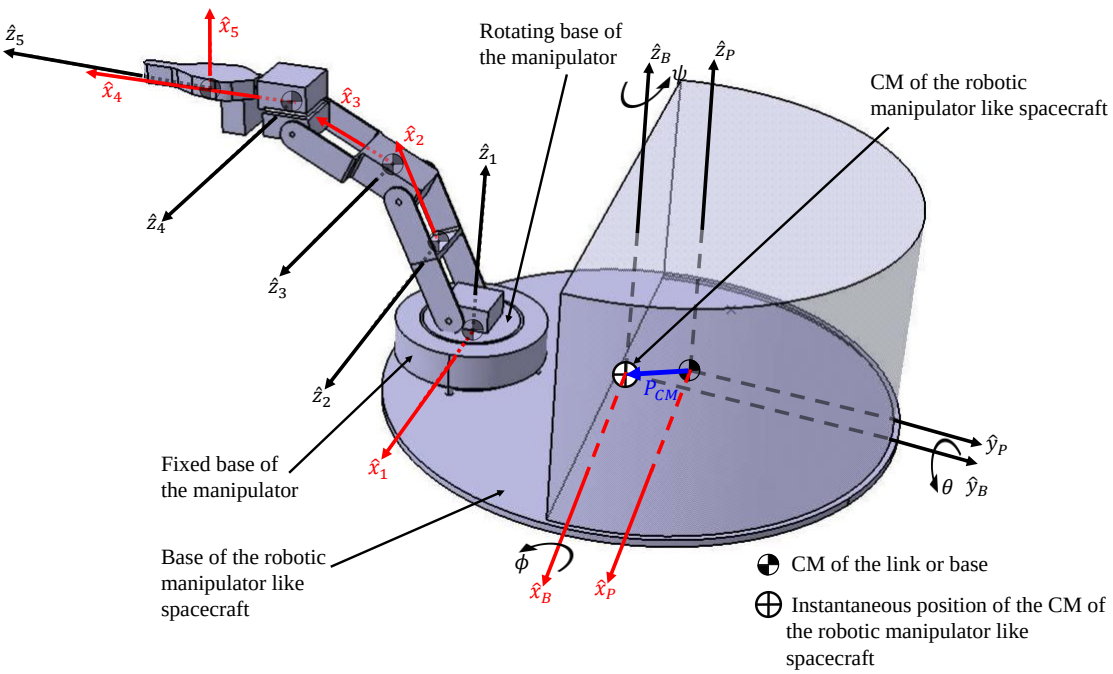
This work deals with dynamics analysis and mathematical modeling of a robotic manipulator-like spacecraft. The mathematical model is obtained by using the Lagrangian formulation including the Lagrange multipliers and the Rayleigh dissipative function to derive the equations of motion from the Hamilton's principle. The Lagrangian Mechanics associated with the Lagrange Multipliers yields the components of the reaction torque/forces acting at the joints which are transmitted from joint to joint up to the robotic platform. The dissipative functions are associated with in the joints motion of the robotic system. In the microgravity environment the interest lays in the fact that the reaction torque/forces affect the motion of the robot platform center of mass and the platform rotational motion as well. The change in the initial center of mass affects the efficiency of the manipulator in accomplishing safely the capture of the target. This paper focuses on the formulation of the equations of motion taking those effects and the computer simulation to analyze the transient phase when the arm is deployed in the direction of the target. The hill's equations are also derived for the computer simulations of the dynamics aiming the control of the relative distance and the relative velocity during the final approach. The attitude dynamics and control during the manipulator operation is object of computer simulations aiming analysis of the interaction between the robot manipulator link motions and the attitude motion control. The equations of motion are computationally simulated by using the Matlab/Simulink® software package. Two control techniques are used, the LQR for the orbital linear motion and the PID (Proportional – Integral – Derivative) aiming the attitude stabilization of the system during the operations of the manipulator.
Key-words - Rendezvous and docking/berthing, manipulator-like spacecraft, on-orbit servicing, Lagrangian, Lagrange's multiplier.

1. Introduction

The rendezvous and docking (RVD) of spacecraft has been subject of studies and tests since the years that preceded the landing of the men on the moon. The orbital maneuvers included astronauts commanding the orbital operations of RVD. However, new technologies of information and new hardware (sensors and actuators) as well software development brought about a new era of applications of the orbital operation of rendezvous and docking/berthing (RVD/B). The possibility of on-orbit servicing (OOS) such as spacecraft maintenance, refueling, repairing, rescuing, and inserting satellite into orbits, taking/exchanging astronauts in orbital space stations, re-supplying the orbital stations (food, water, fuel), cleaning the orbital environment by removing debris, assembling large space structures, retrieval of spacecraft (capture and return to ground), rejoining an orbiting vehicle to land in planetary exploration, opened new areas of study of RVD/B orbital operations.

The RVD/B process consists of a series of orbital maneuvers and controlled trajectories, which successively bring the chaser spacecraft into the vicinity of the target and dock or berth with it. The final approach trajectory must put the chaser spacecraft inside the narrow boundaries of position, velocities, attitude, and angular rates required for the docking or capture process. In case of berthing the chaser spacecraft guidance navigation and control (GNC) subsystem guides the vehicle to a nominally zero relative velocity and angular rates inside the workspace of a robotic manipulator. The manipulator may be located either on the chaser or target vehicle.

The manipulator captures it and guides it to the final position and then inserts it into the interfaces of the relevant target berthing port [Fehse, 2003]. This berthing process is safer than the direct docking process since it prevent collision. The main phase of the RVD/B orbital maneuvers are given in [Fehse, 2003] and are summarized as launch, phasing, far range rendezvous, close range rendezvous and mating. In this paper it is assumed that the robotic manipulator-like spacecraft has reached the close range rendezvous phase successfully, and is ready to accomplish the docking or berthing.
From the launch to the phasing phase the navigation is absolute, that is, it is based on absolute measurements in an Earth-centered frame. The next phases use the relative navigation and accordingly the relative motion is considered in the in this paper. The relative motion refers to the motion of the manipulator-like spacecraft center of mass. In other words the motion is analyzed with respect to (w.r.t.) motion of the target or the chaser. In this paper a non-



cooperative target is assumed and then the motion is described w.r.t. the chaser centered fixed

Fig. 1- Physical Model for the Manipulator-like Spacecraft and frames

frame, in coordinates of the orbital frame, shown if figure 2. This last phase can be divided into two sub phases, the capture and the structural connection. In this paper the structural connection is not considered and only the transient phase is considered, when the manipulator-like spacecraft captures the target. Figure 1 shows the concept of a manipulator-like spacecraft used in this work and several frames defined in the arm joints and a frame defined in the instantaneous cm of mass of the manipulator-like spacecraft. The frame $\hat{x}_p\hat{y}_p\hat{z}_p$ is fixed in the platform and follows it in its attitude motion w.r.t. the orbital frame, $\hat{x}_{lvlh}\hat{y}_{lvlh}\hat{z}_{lvlh}$, (figure 2).

2. Mathematical modeling approach by using the Lagrangian formulation with the Lagrange's multipliers

The physical model adopted for the sake of the derivation of the equations of motions is shown in figure 1. It comprises a platform containing the robot manipulator and the system is a manipulator-like spacecraft. Figure 1 shows several frames. The subscript *b* means body axes with origin in the instantaneous center o mass, CM. The subscript *p* means platform and the associated frame (platform frame) is parallel to the body frame with origin in the instantaneous CM. The attitude equations of motion are described in the platform frame in terms of the orbital

LVLH (Local vertical, Local horizontal) frame coordinates, $\hat{x}_{lvlh}\hat{y}_{lvlh}\hat{z}_{lvlh}$ (shown in figure 2). The origin of this frame is the instantaneous CM. The frames $\hat{x}_i\hat{y}_i\hat{z}_i$ i = 1, 2 … 5 are defined in the joints of the manipulator by following the Denavit-Hartenberg convention as given in Craig's book [Craig, 2005]. Figure 1 also illustrates the attitude angles indicated as rotations about the instantaneous body axes defined by $\boldsymbol{P}_{\mathrm{CM}}$ vector shown in Fig. 1. The impact of changes in center of mass is discussed in [Skkar, 1994]. The in-orbit configuration and frames are shown in figure 2 for the orbital motion. The orbital motion refers to the CM motion of the manipulator-like spacecraft around the Earth. Note that both the LVLH and the body $\hat{x}_B\hat{y}_B\hat{z}_B$ frames have their origin at the same point, are fixed in the orbit and moves around the Earth. However, the body frame also rotates w.r.t. the orbital LVLH (attitude and attitude rates). Both frames are related to each other through a rotation matrix or attitude matrix. Ω, ω, i, and f are the keplerian elements describing the orbital motion. The vernal equinox or First Point of Aries refers to the inertial direction of X (See figure 3). The equatorial plane contains X and Y axes, normal to each other while Z axis is normal to the equator plane completing the frame definition according to the right-hand rule. The $\hat{z}_{lvlh}$ follows the local vertical (Nadir direction) and rotates synchronized with the orbit. It is the yaw axis. $\hat{x}_{lvlh}$ is normal to Nadir and points toward the velocity vector. $\hat{y}_{lvlh}$ is the pitch axis in the negative orbit normal direction. In a nominal configuration both $\hat{x}_{lvlh}\hat{y}_{lvlh}\hat{z}_{lvlh}$ and the LVLH are aligned. The $\hat{x}_{lvlh}\hat{y}_{lvlh}\hat{z}_{lvlh}$ frame is convenient for position and orientation of spacecraft instruments [Wertz, 1987].

2.1. The orbital relative motion for RVD in close range approach

In this paper it is considered the close range approach successfully reached by the chaser spacecraft. The analysis is focused in the final approach and capture of the target considering the control of the relative velocity of the CM and the attitude control. Such proximity operation problem of two space vehicles is of primary importance to many present and planned space missions. In this final approach the linear equation for the orbital motion is suitable for the dynamic analysis. Such equation are frequently called Cloresshy-Wiltshire or Hill's equations. This paper considers the target as seen from the chaser. This is to say that the equations of motion are written w.r.t. to the chaser's frame. The details of the derivation of hill's equation for the linear system equations can be seen in [Wie, 2008 and Fhese, 2003]. The Hill's equations are

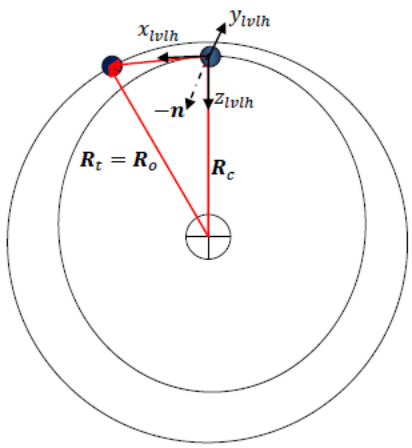


Fig. 2 In-orbit chaser and target configuration

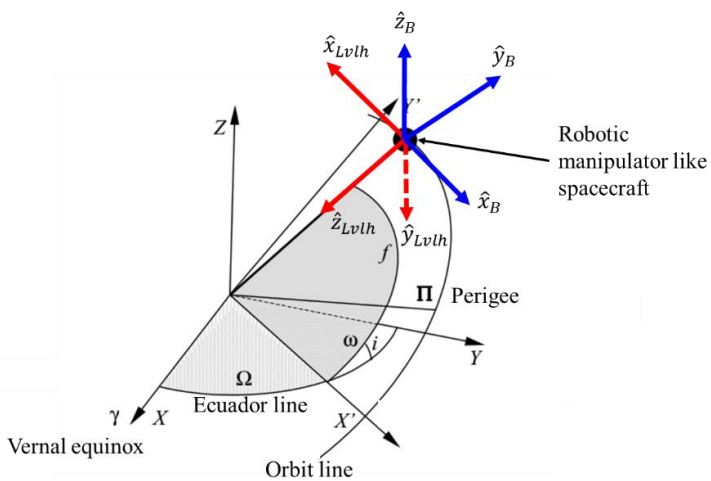


Fig. 3 - Inertial, LVLH, and body frames

presented in the next. Thes equations are obtained taking into account the assumption that the target is on an circular orbit. The Hill's equations are

$$\begin{aligned} &\ddot{x} - 2n\dot{z} = f_x \\ &\ddot{y} + 2n^2 y = f_y \\ &\ddot{z} - 2nx - 3n^2 z = f_z \end{aligned} \tag{1}$$

where n is the orbital rate.
This system Linear Quadratic Regulator (LQR) technique is used for the sake of control of the translational motion. To implement the control formulation consider the state vector associated with the equations of dynamics

$$\{X\} = \begin{Bmatrix} x \\ y \\ z \\ \dot{x} \\ \dot{y} \\ \dot{z} \end{Bmatrix} = \begin{Bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \\ x_6 \end{Bmatrix} \tag{2}$$

The state equations can be written as

$$\begin{Bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \\ \dot{x}_4 \\ \dot{x}_5 \\ \dot{x}_6 \end{Bmatrix} = \begin{bmatrix} 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 2n \\ 0 & n^2 & 0 & 0 & 0 & 0 \\ 0 & 0 & -3n^2 & 2n & 0 & 0 \end{bmatrix} \begin{Bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \\ x_6 \end{Bmatrix} + \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ \frac{1}{m_c} & 0 & 0 \\ 0 & \frac{1}{m_c} & 0 \\ 0 & 0 & \frac{1}{m_c} \end{bmatrix} \begin{Bmatrix} u_1 \\ u_2 \\ u_3 \end{Bmatrix} \tag{3}$$

The LQR technique can be stated as

Minimize a cost function given by

$$J = \int_0^{\propto} \{X\}^T [Q]\{X\} + \{u\}^T [Q]\{u\}\, dt \tag{4}$$

Subject to

$$\{\dot{X}\} = [A]\{X\} + [B]\{u\} \tag{5}$$

$$\{u\} = -[K]\{X\} \tag{6}$$

where $\{u\}$ is the state feedback control law that results of the quadratic performance index optimization subjected to the state equation. [A] and [B] are the plant and input matrices, respectively. [Q] is a positive semi definite state weighting matrix, and [R] is the control weighting matrix that has to be positive definite. [Q] is *2n×2n* and [R] is *p×p.* This technique requires the solution of the Riccati algebraic equation given by

$$[A]^T[P] + [P][A] + [Q] - [P][B][R]^{-1}[B]^T[P] = 0 \tag{7}$$

where $[P]$ is the Riccati matrix. This control technique is implemented in the MATLAB® software environment.

2.2. Equations of the attitude dynamics of the manipulator-like spacecraft

The Lagrange's multipliers are used to derive the equations of motion from the Hamilton's principle. The Lagrange's formula includes also the Rayleigh's dissipative function *F* in approaching the energy dissipation [Meirovitch, L] inherent to the manipulators' joints. The resulting Lagrange's formula including the Lagrange's multiplier cam be written as

$$\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{q}_k}\right) - \frac{\partial L}{\partial q_k} = Q_k + Q'_k = Q_k + \sum_{\ell=1}^{m} \lambda_{\ell k} a_{\ell k} \tag{8}$$

where $Q'_k = \sum_{\ell=1}^{m} \lambda_{\ell k} a_{\ell k}$, $k = 1, 2 \ldots, n$ and $\ell = 1,2,..,m$ constraints where n are the system degree of freedom considered in the problem. Assuming that all the forces derived from a potential are accounted for in the Lagrangian *L*, the Q_k represents all non conservative forces acting on the system. In the derivation of the equations for the manipulator-like spacecraft it is considered only the dissipative forces acting in manipulator joints, so the Lagrange's formula to approach the problem can be written as

$$\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{q}_k}\right) - \frac{\partial L}{\partial q_k} + \frac{\partial F}{\partial \dot{q}_k} = \sum_{\ell=1}^{m} \lambda_{\ell k} a_{\ell k} \tag{9}$$

which takes into account the Rayleigh dissipative function *F* accounted for in the generalized forces Q_k . The term in the right-side of the equation represents the constraint forces, Q'_k. These forces are important for robot manipulators dynamic analysis because they refer to the joint reaction forces that propagate from link to link of the robot arm up to platform. As the system environment is characterized by the microgravity those reactions forces cause stability problems and also affect the efficiency of the manipulator performance. *L* is the Lagrangian function defined as

$$L = T - V \tag{10}$$

where *T* and *V* are the kinetic and potential energies. If no force arising from a potential is considered then the Lagrangian function reduce to the kinetic energy, as it is the case approached in this paper. The relative linear equations (Hill's equations) which include gravitational forces were derived separately by using the Newton's law of gravitation.
The formulation of the equations of motion for a manipulator requires the computation and manipulation of the homogeneous transformations matrices that include the rotational and translational motion of the links of the manipulator. The algebraic manipulations become very hard as the number of links increases. The details of obtaining and manipulating the transformations matrices from joint to joint of the manipulator as well as the kinematic equations are shown in the Craig's book [Craig, 2005]. By assuming five joints and a platform as shown in the figure 1, the kinetic energy can be derived for each part of the manipulator-like spacecraft and then the total energy is obtained by summation of the parts of the system, that is

$$T = T_p + \sum_{i=1}^{5} T_i \tag{11}$$

Where the subscript p means platform and *i* = 1, 2, .. , 5 are the links of the manipulator.
In order to derive the equations of motion for the manipulator-like spacecraft it is required to derive the inertia matrices of the platform and the inertia matrices for each of the manipulator

links. All the inertias matrices must be transferred to the platform frame if the equations of motion are to be written in that frame. The Steiner's theorem (parallel axis theorem) and rotation transformations matrices are used to write the total inertia matrix in platform frame. Alternatively, the equations can be written w.r.t. in t the end-effector frame. This can be done by using the homogeneous transformation from the link frame to link frame up to the end-effector frame [Craig, 2005 and Greenwood, 1987].
Considering the Jacobian concept [Bong Wie, 2008 and Craig, 2005] the linear and angular rates can be written as

$$\begin{Bmatrix} v \\ \omega \end{Bmatrix} = [J(q)]\{\dot{q}\} \tag{12}$$

This equation can be split in the form [Craig, 2005]

$$\{v\} = [J_v(q)]\{\dot{q}\} \tag{13}$$

$$\{\omega\} = [J_\omega(q)]\{\dot{q}\} \tag{14}$$

The Jacobian is the matrix of the partial derivatives of the translational velocities as well as the angular velocities.
If it is to represent the velocities in different frames than the use of transformation matrices shall be used. Using this approach the kinetic energy for each part of the robotic manipulator can be written as

$$T_p = \frac{1}{2} m_p \{\dot{q}\}^T \left[J_{v_p}(q)\right]^T \left[J_{v_p}(q)\right] \{\dot{q}\} + \frac{1}{2}\{\dot{q}\}^T \left[J_{\omega_p}(q)\right]^T [{}^{lv}_{p}R]^T [{}^{lv}_{p}I][{}^{lv}_{p}R] \left[J_{\omega_p}(q)\right] \{\dot{q}\} \tag{15}$$

$$T_1 = \frac{1}{2} m_1 \{\dot{q}\}^T \left[J_{v_1}(q)\right]^T \left[J_{v_1}(q)\right] \{\dot{q}\} + \frac{1}{2}\{\dot{q}\}^T \left[J_{\omega_1}(q)\right]^T [{}^{lv}_{1}R]^T [{}^{lv}_{1}I][{}^{lv}_{1}R] \left[J_{\omega_1}(q)\right] \{\dot{q}\} \tag{16}$$

$$T_2 = \frac{1}{2} m_2 \{\dot{q}\}^T \left[J_{v_2}(q)\right]^T \left[J_{v_2}(q)\right] \{\dot{q}\} + \frac{1}{2}\{\dot{q}\}^T \left[J_{\omega_2}(q)\right]^T [{}^{lv}_{2}R]^T [{}^{lv}_{2}I][{}^{lv}_{2}R] \left[J_{\omega_2}(q)\right] \{\dot{q}\} \tag{17}$$

$$T_3 = \frac{1}{2} m_3 \{\dot{q}\}^T \left[J_{v_3}(q)\right]^T \left[J_{v_3}(q)\right] \{\dot{q}\} + \frac{1}{2}\{\dot{q}\}^T \left[J_{\omega_3}(q)\right]^T [{}^{lv}_{3}R]^T [{}^{lv}_{3}I][{}^{lv}_{3}R] \left[J_{\omega_3}(q)\right] \{\dot{q}\} \tag{18}$$

$$T_4 = \frac{1}{2} m_4 \{\dot{q}\}^T \left[J_{v_4}(q)\right]^T \left[J_{v_4}(q)\right] \{\dot{q}\} + \frac{1}{2}\{\dot{q}\}^T \left[J_{\omega_4}(q)\right]^T [{}^{lv}_{4}R]^T [{}^{lv}_{4}I][{}^{lv}_{4}R] \left[J_{\omega_4}(q)\right] \{\dot{q}\} \tag{19}$$

$$T_5 = \frac{1}{2} m_5 \{\dot{q}\}^T \left[J_{v_5}(q)\right]^T \left[J_{v_5}(q)\right] \{\dot{q}\} + \frac{1}{2}\{\dot{q}\}^T \left[J_{\omega_5}(q)\right]^T [{}^{lv}_{5}R]^T [{}^{lv}_{5}I][{}^{lv}_{5}R] \left[J_{\omega_5}(q)\right] \{\dot{q}\} \tag{20}$$

The total kinetic energy can be written as

$$T = \frac{1}{2}\{\dot{q}\}^T \Big(m_p \left[J_{v_p}(q)\right]^T \left[J_{v_p}(q)\right] + \left[J_{\omega_p}(q)\right]^T [{}^{lv}_{p}R]^T [{}^{lv}_{p}I][{}^{lv}_{p}R] \left[J_{\omega_p}(q)\right] + \sum_{\ell=1}^{5} m_\ell \left[J_{v_\ell}(q)\right]^T \left[J_{v_\ell}(q)\right] + \{\dot{q}\}^T \left[J_{\omega_\ell}(q)\right]^T [{}^{lv}_{\ell}R]^T [{}^{lv}_{\ell}I][{}^{lv}_{\ell}R] \left[J_{\omega_\ell}(q)\right] \Big) \{\dot{q}\} \tag{21}$$

Defining

$$[D] = m_p \left[J_{v_p}(q)\right]^T \left[J_{v_p}(q)\right]\{\dot{q}\} + \frac{1}{2}\{\dot{q}\}^T \left[J_{\omega_p}(q)\right]^T \left[{}^{lv}_{p}R\right]^T \left[{}^{lv}_{p}I\right] {}^{lv}_{p}[R] \left[J_{\omega_p}(q)\right] + \sum_{\ell=1}^{5} m_\ell \left[J_{v_\ell}(q)\right]^T \left[J_{v_\ell}(q)\right] + \left[J_{\omega_\ell}(q)\right]^T \left[{}^{lv}_{\ell}R\right]^T {}^{lv}_{\ell}[I] \left[{}^{lv}_{\ell}R\right] \left[J_{\omega_\ell}(q)\right] \quad (22)$$

The kinetic energy can be written in a more compact form a

$$T = \frac{1}{2}\{\dot{q}\}^T [D]\{\dot{q}\} \quad (23)$$

2.3. Constraint equations and the Lagrange's multipliers

A more systematic and general approach for developing the equations of motion taking into account the reaction forces (or the augmented equations of motion) is based on the Lagrangian dynamics. In the Lagrangian approach, the method of Lagrange multipliers is used to define generalized constraint forces and to obtain an augmented set of equations in which the coefficient matrix is symmetric. In order to include the Lagrange's multipliers [Meirovitch, 2003, Shabana, 2010] it is necessary to include the constraints associated with the reaction forces in the joints. The derivation of the constraint equations is shown next. Let the constraint force be written as

$$Q'_k = \sum_{\ell=1}^{m} \lambda_{\ell k} a_{\ell k} \quad (24)$$

where $\lambda_{\ell k}$ are parameters related to the magnitude of the constraint forces and

$$a_{\ell k} = \frac{\partial g_\ell}{\partial q_k} \quad (25)$$

where $g_\ell(q_1, q_2, \dots, q_n, t) = 0$ $\ell = 1, 2, \dots, m$ are the holonomic constrains of the system. Then

$$Q'_k = \sum_{\ell=1}^{m} \lambda_{\ell k} a_{\ell k} = \sum_{\ell=1}^{m} \lambda_{\ell k} \frac{\partial g_\ell}{\partial q_k} \quad (26)$$

Writing Q in matrix notation.

$$\{Q_c\} = -[C_d]^T\{\lambda\} \quad (27)$$

where $\{Q_c\}$ is the vector of the reaction forces, $[C_d]$ is the constrained Jacobian matrix and $\{\lambda\}$ is the vector proportional to the constraint forces.

The constraint coordinates written w.r.t. LVLH frame are 15 positions since it is considered five links to each one is attached a xyz frame. In matrix notation the

$$q_c = [x_{Lv1}\ y_{Lv1}\ z_{Lv1}\ x_{Lv2}\ y_{Lv2}\ z_{Lv2}\ x_{Lv3}\ y_{Lv3}\ z_{Lv3}\ x_{Lv4}\ y_{Lv4}\ z_{Lv4}\ x_{Lv5}\ y_{Lv5}\ z_{Lv5}]^T \qquad (28)$$

These coordinates refer to the position of the CM of each link w.r.t. the LVLH frame. Then the constraint equation may be written as [Unfried, 2017]

$$\{c_1\} = \left(\left\{p_{cm_p}\right\} + {}^{Lv}_{P}[R]\begin{Bmatrix} x_{p1} \\ y_{p1} \\ z_{p1} \end{Bmatrix}\right) - \left(\left\{p_{cm_1}\right\} - {}^{Lv}_{1}[R]\begin{Bmatrix} 0 \\ 0 \\ z_{cm_1} \end{Bmatrix}\right),$$

$$\{c_2\} = \left(\left\{p_{cm_1}\right\} + [{}^{Lv}_{1}R]\begin{Bmatrix} 0 \\ 0 \\ z_{12} - z_{cm_1} \end{Bmatrix}\right) - \left(\left\{p_{cm_2}\right\} - [{}^{Lv}_{2}R]\begin{Bmatrix} x_{cm_2} \\ 0 \\ z_{cm_2} \end{Bmatrix}\right) \qquad (29)$$

$$\{c_3\} = \left(\left\{p_{cm_2}\right\} + [{}^{Lv}_{2}R]\begin{Bmatrix} x_{23} - x_{cm_2} \\ 0 \\ -z_{cm_2} \end{Bmatrix}\right) - \left(\left\{p_{cm_3}\right\} - [{}^{Lv}_{3}R]\begin{Bmatrix} x_{cm_3} \\ 0 \\ z_{cm_3} \end{Bmatrix}\right),$$

$$\{c_4\} = \left(\left\{p_{cm_3}\right\} + [{}^{Lv}_{3}R]\begin{Bmatrix} x_{34} - x_{cm_3} \\ 0 \\ -z_{cm_3} \end{Bmatrix}\right) - \left(\left\{p_{cm_4}\right\} - [{}^{Lv}_{4}R]\begin{Bmatrix} x_{cm_4} \\ y_{cm_4} \\ z_{cm_4} \end{Bmatrix}\right),$$

$$\{c_5\} = \left(\left\{p_{cm_4}\right\} + [{}^{Lv}_{4}R]\begin{Bmatrix} x_{45} - x_{cm_4} \\ y_{45} - y_{cm_4} \\ 0 \end{Bmatrix}\right) - \left(\left\{p_{cm_5}\right\} - [{}^{Lv}_{5}R]\begin{Bmatrix} x_{cm_5} \\ y_{cm_5} \\ z_{cm_5} \end{Bmatrix}\right),$$

The vector of constraint equations can be written in the matrix notation as

$$\{\boldsymbol{C}(\boldsymbol{q})\} = \begin{Bmatrix} c_1 \\ c_2 \\ c_3 \\ c_4 \\ c_5 \end{Bmatrix} = 0. \qquad (30)$$

Applying the Lagrange's formula given by Eq.(9) for $L = T$ (No forces derived from potential is considered in the Lagrangian function), yields

$$\left\{\frac{\partial T}{\partial \dot{q}_k}\right\} = \frac{1}{2}\frac{\partial}{\partial \dot{q}_k}(\{\dot{q}\}^T[D(q)]\{\dot{q}\}) = [D(q)]\{\dot{q}\},\ k = 1, 2, \dots, 23, \qquad (31)$$

$$\left\{\frac{d}{dt}\left(\frac{\partial T}{\partial \dot{q}_k}\right)\right\} = \left[\dot{D}(q)\right]\{\dot{q}\} + [D(q)]\{\ddot{q}\},\ k = 1, 2, \dots, 23, \qquad (32)$$

$$\left\{\frac{\partial T}{\partial q_k}\right\} = \frac{1}{2}\frac{\partial}{\partial q_k}(\{\dot{q}\}^T[D(q)]\{\dot{q}\}) = \frac{1}{2}\{\dot{q}\}^T\frac{\partial[D(q)]}{\partial q_k}\{\dot{q}\},\ k = 1, 2, \dots, 23. \qquad (33)$$

$$\left\{\frac{\partial F}{\partial \dot{q}_k}\right\} = \frac{1}{2}\frac{\partial}{\partial \dot{q}_k}\left(\{\dot{q}\}^T[c]\{\dot{q}\}\right) = \{c\}\cdot\{\dot{q}\} = \{c\}\,[I_d]\,\{\dot{q}\},\ \ k = 1, 2, \dots, 23, \tag{34}$$

So the equations of motions are

$$[D(q)]\{\ddot{q}\} + \left[\dot{D}(q)\right]\{\dot{q}\} - \frac{1}{2}\dot{q}^T\left[\frac{\partial D(q)}{\partial q_k}\right]\{\dot{q}\} + [c]\{\dot{q}\} = \{Q\} - \left[C_q\right]^T\{\lambda\},\ \ k = 1, \dots, 23. \tag{35}$$

where [c] is a matrix containing the damping coefficients associated to the Rayleigh damping function.
Considering that the holonomic constraints for the problem approached here do not depend explicitly on time, the first and second derivates of Eq. (30) yields

$$\left[C_q\right]\{\dot{q}\} + C_t = 0 \tag{36}$$

As the vector of constraints equations $\{C(q,t)\} = \{C(q)\} = 0$, i.e. the constraints do not depend explicitly on time, its first and second derivatives w.r.t. time can be written as:

$$\frac{d}{dt}\left(\left[C_q\right]\{\dot{q}\} + C_t\right) = 0 \tag{37}$$

Using the chain rule of differentiation, the time derivative of Eq. (36) yields

$$\left[C_q\right]\{\ddot{q}\} + \left[\left(\left[C_q\right]\{\dot{q}\}\right)_q\right]\{\dot{q}\} \tag{38}$$

By defining

$$[Q_d] = -\left[\left(\left[C_q\right]\{\dot{q}\}\right)_q\right]\{\dot{q}\} \tag{39}$$

The Eq.(38) can then be written in a compact form as (40)

$$\left[C_q\right]\{\ddot{q}\} = \{Q_d\}$$

Where the $\{C(q,t)\}$ is a vector whose components are the constraints. $\left[C_q\right]$ is the system constraint Jacobian matrix. $\{C_t\}$ is a vector of the partial derivatives of the constraints w.r.t. time (indicated by the subscript *t*) and has the general form [Shabana, 2011]
$\{C_t\} = \left[\frac{\partial C_1}{\partial t}\ \ \frac{\partial C_2}{\partial t}\ \dots\ \frac{\partial C_n}{\partial t}\right]^T$. $\left[\left(\left[C_q\right]\{\dot{q}\}\right)_q\right]$ is a square matrix in the position and velocities of the joints with respect to the LVLH frame.

The Eq.(35) can be written in a more compact form as

$$[M']\{\ddot{q}\} = \{Q\} - [M_v]\{\dot{q}\} + \frac{1}{2}\{\dot{q}\}^T[M_{cc}]\{\dot{q}\} - [M_{rd}]\{\dot{q}\} - \left[C_q\right]^T\{\lambda\}, \tag{41}$$

where

$$[M'] = [D(q)],$$

$$[M_v] = \left[\dot{D}(q)\right],$$

$$[M_{cc}] = \left[\frac{\partial D(q)}{\partial q_k}\right],$$

$$[M_{rd}] = [c]$$

In order to be able to solve the system of equations, it is necessary to have 15 additional equations, which are the nonlinear algebraic constraint equations, as described by Eq. (30). The augmented system of equations can then be written as

$$\begin{bmatrix}\ddot{q}\\ \lambda\end{bmatrix} = \begin{bmatrix}[M'] & [C_q^T]\\ [C_q] & [0]\end{bmatrix}^{-1}\left\{\begin{Bmatrix}Q\\ Q_d\end{Bmatrix} - \begin{bmatrix}[M_v]+[M_{rd}]\\ [0]\end{bmatrix}\{\dot{q}\} + \{\dot{q}\}^T\begin{bmatrix}\frac{1}{2}M_{cc}\\ 0\end{bmatrix}\{\dot{q}\}\right\}, \tag{42}$$

However, because of the complexity of the augmented equations describing the dynamics of the system, the numerical solution of the set of constraint equations is not a straightforward problem. There are some numerical stabilization issues at the constraints. The algebraic solution can be numerically unstable for certain cases because of the so called drift-off effects which consist of the violation of constraints while implementing the numerical solution. It is influenced by round-off errors and discretization due to the numerical approximation. The violation problem can be seen from the fact that numerical solution using the second derivative of the constraints is not the same as the solution using the original constraints equations. Due to its simplicity and easiness for computational implementation, the Baumgarte's method [Shabana, 2010] is used in this work for the constraint stabilization. By using the method, Eq.(38) is modified according to

$$\{\ddot{C}\} + 2\alpha\left\{\frac{\partial C}{\partial q}\right\} + (\beta)^2\{C\} = 0, \tag{43}$$

where α and β are paramenters to be chosen according to $\alpha > 0$ and $\beta \neq 0$. Then Eq. (37) is redefined in terms of Eq (42) as

$$[C_q]\{\ddot{q}\} = \{Q_d\} - 2\alpha\left([C_q]\{\dot{q}\}\right) - (\beta)^2\{C\}. \tag{44}$$

Then Eq.(41) can be rewritten as

$$\begin{bmatrix}\ddot{q}\\ \lambda\end{bmatrix} = \begin{bmatrix}M' & C_q^T\\ C_q & 0\end{bmatrix}^{-1}\left(\begin{bmatrix}Q\\ \{Q_d\} - 2\alpha\left([C_q]\{\dot{q}\}\right) - (\beta)^2\{C\}\end{bmatrix} - \begin{bmatrix}[M_v]+[M_{rd}]\\ 0\end{bmatrix}\{\dot{q}\} + \{\dot{q}\}^T\begin{bmatrix}\frac{1}{2}M_{cc}\\ 0\end{bmatrix}\{\dot{q}\}\right), \tag{45}$$

where α and β are the Baumgarte's parameters. From [Flores, 2008] as an initial guess for α and β for this problem, it can be used:

$$\alpha = \frac{1}{t_s},$$

$$\beta = \frac{\sqrt{2}}{t_s}, \tag{46}$$

where t_s is the integration time step.
This approach to find the values of α and β works fine for cases where the time step is not too small which is not the case of this work. For the problem under analysis, is necessary to work with a small time step because of the complexity of the dynamics. From this, the values of the

parameters found from Eqs.(43) can be used only as an initial guess. It is important to highlight that, when using the stabilization method, it is necessary to keep a fixed time step.

3. Computer simulations and results

In order to implement the dynamic analysis the linear orbital relative motion between the manipulator-like spacecraft and the target as well as the attitude dynamics equations of motion are simulated in computer by using the MATLAB® and Simulink software environment. Two different control techniques are implemented: the LQR and the PID. The LQR is more consistent for linear system equations as it is the case of the relative orbital motion in the final approach phase of RVD. The PID is used for maneuvering the manipulator links during the capture of the target whose motion is not restricted to small angles. The main goal of the dynamics analysis is to compute the control effort of the actuators at the manipulator joints without controlling the attitude, the control effort to control the motion of the manipulator-like spacecraft while the manipulator is being under control and the attitude is also simultaneously under control, the effect of the joint reaction torque on the attitude and control efforts.

The control algorithms are to be subjected of lab test validation and as well as the results of these simulations. For consistency with future test validations the mass properties (inertias and center o masses definition) used for the computer simulations shown here are those (table 1) of two existing floating robots at ITA. The floating robot physical models are shown in figure 1 and the attitude equations of motion were obtained for that model by considering it in-orbit configuration as shown in figure 2. Fist it is shown the results for the linear close range relative motion. Two aspects are mandatory to be analyzed, the final relative distance and relative velocity. The relative distance must be consistent with the workspace of the manipulator-like spacecraft otherwise the capture maneuvers fail to grasp the target. The relative velocity must be zero to avoid collision of both spacecraft during the capture. It is

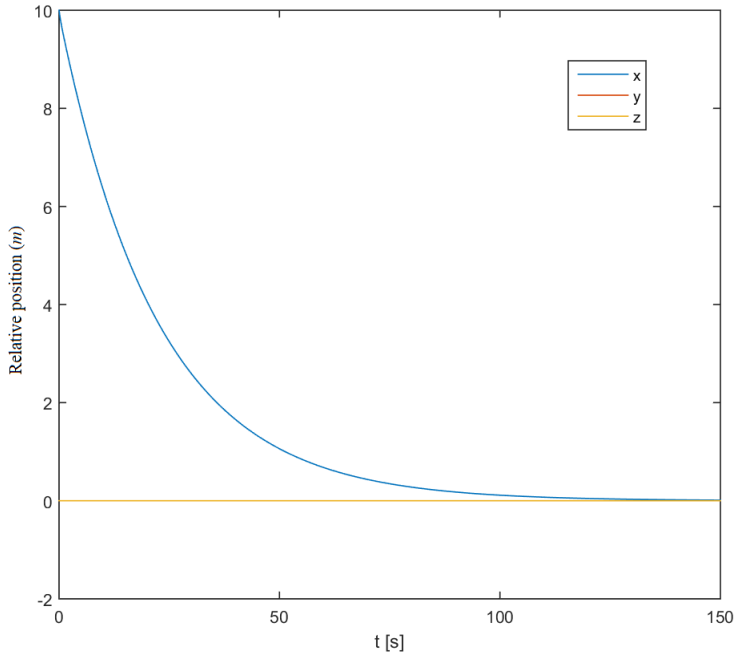


Fig. 4- Relative position versus time

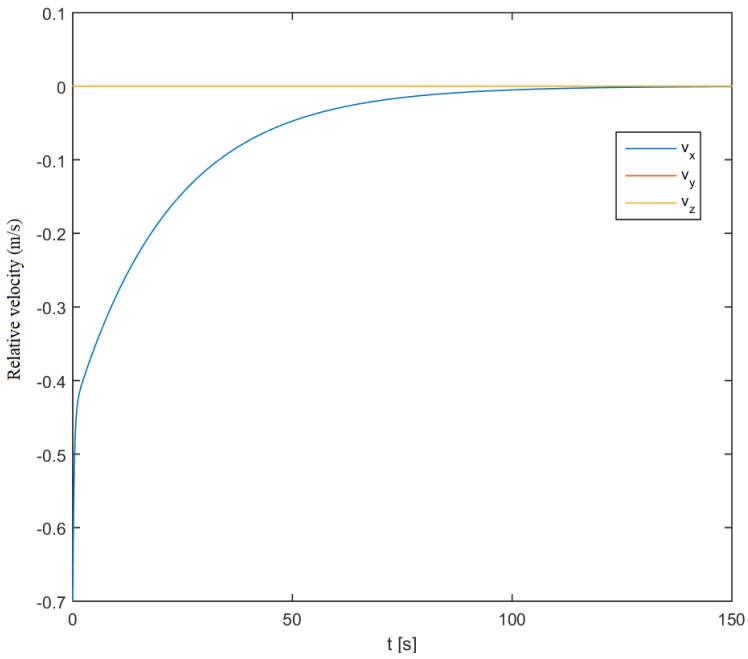


Fig. 5 - Relative velocity versus time

assumed in this work that the attitude of the target is known as well as the inertia properties of the target. The manipulator-like spacecraft shall track that target attitude and synchronize with it just before the capture is implemented. Figure 4 and 5 show the controlled close approach relative position and velocity, respectively.
It is mandatory to synchronize the linear motion between the manipulator-like spacecraft so as the relative velocity between them be zero. zero, the grasping operation may fail.
The input inertia matrices for the attitude dynamics and control simulations are given in Table 1.

Table 1 - Input inertia matrices for the computer simulations

Platform	Frame 1	Frame 2	Frame 3
$Ixx_P = 0.268;$ $Iyy_P = 0.285;$ $Izz_P = 0.251;$ $Ixy_P = -3.383e\text{-}5;$ $Ixz_P = -8.165e\text{-}7;$ $Iyz_P = -0.084;$	$Ixx_1 = 7.547e\text{-}5;$ $Iyy_1 = 7.539e\text{-}5;$ $Izz_1 = 9.299e\text{-}5;$ $Ixy_1 = -8.291e\text{-}8;$ $Ixz_1 = 5.427e\text{-}7;$ $Iyz_1 = 1.658e\text{-}7;$	$Ixx_2 = 2.867e\text{-}5;$ $Iyy_2 = 1.998e\text{-}4;$ $Izz_2 = 1.757e\text{-}4;$ $Ixy_2 = 9.898e\text{-}22;$ $Ixz_2 = 4.86e\text{-}6;$ $Iyz_2 = -3.666e\text{-}23;$	$Ixx_3 = 3.242e\text{-}5;$ $Iyy_3 = 1.637e\text{-}4;$ $Izz_3 = 1.434e\text{-}4;$ $Ixy3 = -1.577e\text{-}25;$ $Ixz_3 = 1.097e\text{-}5;$ $Iyz_3 = 3.308e\text{-}26;$

Frame 4	Frame 5
$Ixx_4 = 1e\text{-}4;$ $Iyy_4 = 1.024e\text{-}4;$ $Izz_4 = 1.106e\text{-}4;$ $Ixy_4 = 3.426e\text{-}5;$ $Ixz_4 = 2.616e\text{-}5;$ $Iyz_4 = 2.989e\text{-}5;$	$Ixx_5 = 2.37e\text{-}4;$ $Iyy_5 = 2.388e\text{-}4;$ $Izz_5 = 1.002e\text{-}4;$ $Ixy_5 = 3.941e\text{-}5;$ $Ixz_5 = 3.942e\text{-}5;$ $Iyz_5 = -3.667e\text{-}5;$

In order to test the correctness of the equations of motion it was implemented a computer simulation with attitude control disabled and the robotic arm in its nominal initial configuration (not moving). No environment perturbation is considered in the mathematical model, that is, no external forces and torques were included. Considering such dynamics scenario no changes in the center of mass position of the manipulator-like spacecraft should occur. It was expected results of no control effort, no change in the robot arm position, no change in the manipulator center of mass position, since without external perturbation and control actions the configuration in attitude and center of mass should not change. The simulation of this scenario confirmed that expect stable motion.
Then the computer simulations in which the robot arms are moving, i.e., the manipulator is operating. Also it is considered that attitude control is disabled. In this scenario the actuators act on the joints of the manipulator changing the mass configuration of the system as the robot arm moves. For this case it was expected that the arm motion affect the attitude motion due to the reactions on the joints that propagates from link to link up to the manipulator platform. The

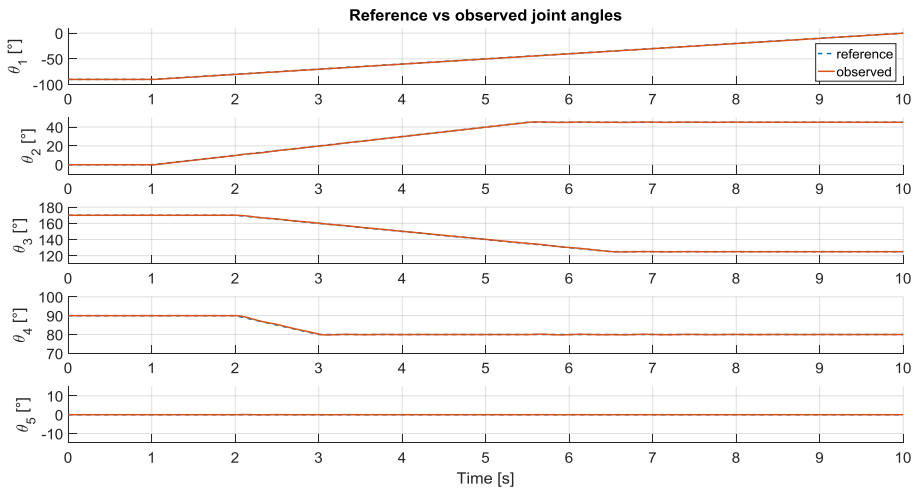


Fig. 6 The joints of the robot arm moves

results in Figure 6 to 9 highlight the expected results. Figure 6-7 shows the joint angles position w.r.t. time and the associate control effort to move the joint angles. The evaluation of the control effort is important to in the specification and acquiring of appropriate actuators.

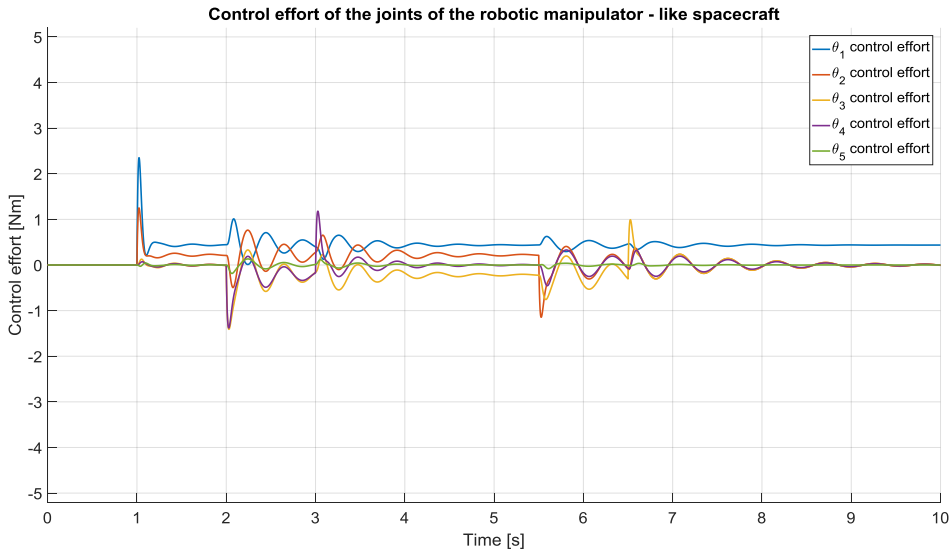


Fig. 7 – Control effort of each joint actuator do move the associated arm link angle

Figure 8 highlights the perturbation on attitude caused by the motion of the arm links. Note that the attitude control is disabled and it is clear the attitude angles deviation from the original position under impact of the control actuation on the joints. The computer simulation for this case considers the actuators acting only on the joints of the arm and because no attitude control is implemented, the perturbation on attitude due to the joint reaction torque is enhanced. The control effort in attitude remains 0. Such result gives a good indication about the correctness of the mathematical model.

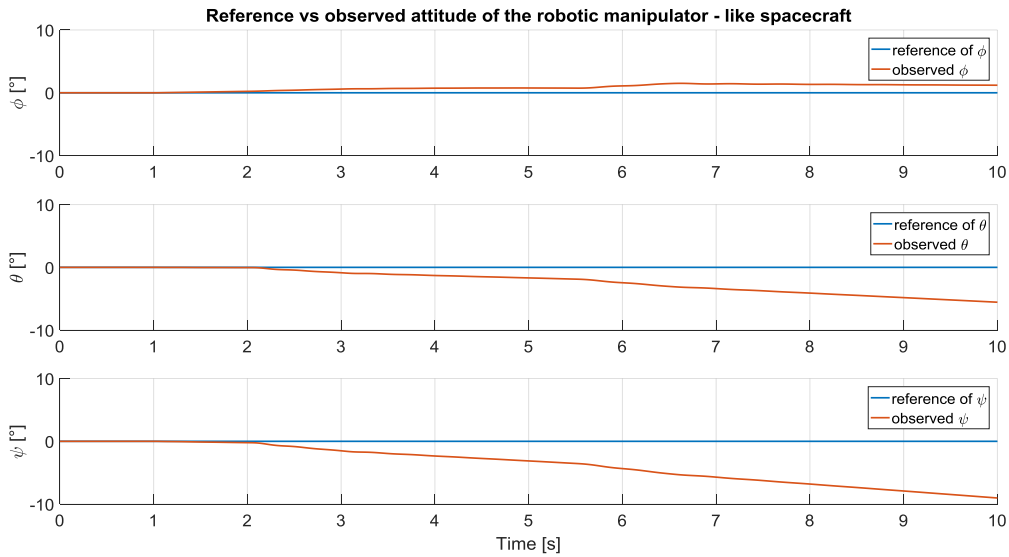


Fig. 8 - Perturbation in attitude caused by the motion of the robot arm

Also the center of mass change w.r.t. time since the motion of the robot arm redistribute masses about the original center of mass, changing it to a new position. This is shown in figure 9.

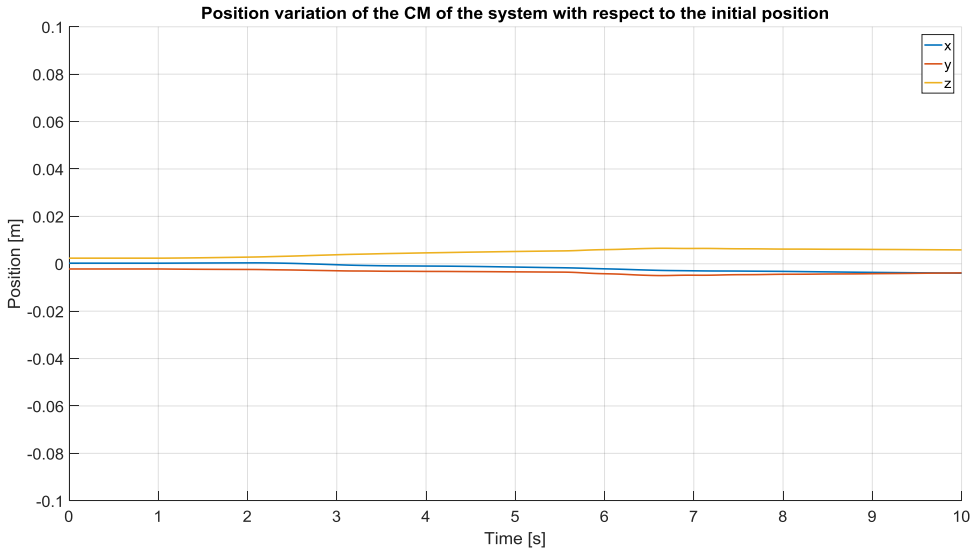


Fig. 9 – Expected changes in the position of the center of mass of the manipulator

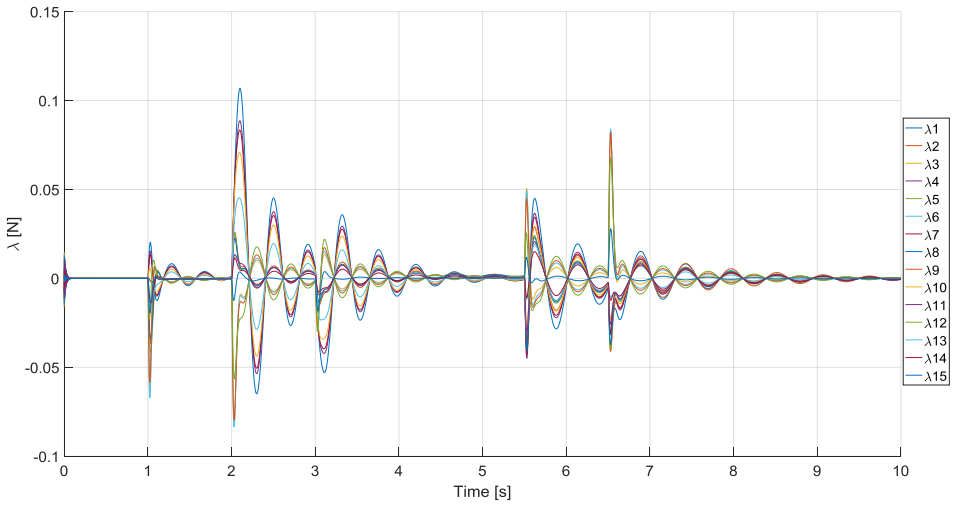


Fig. 10– Reaction forces/torque on the joints

This result is very important due to the high precision required for successful capture operation. If the motion of the center of mass causes misalignment w.r.t. the actuator force direction then a torque is generated risking also the docking operation. The solution is to keep the robotic platform under control to avoid risking the capture operation.

Figure10 provides information about the reactions forces generated in the robot joints. The reactions disturb the attitude and affect the liner motion of the center of mass. The reaction effects appear in these computer simulations because the attitude control is not active. At this point it must be remembered that the Lagrange's multipliers are related to the reaction forces which affect center of mass behavior.

Figures (11-13) are generated by the following dynamics configuration, the control is active and the arm is moving. The dynamics analysis is focused on the case in which the attitude is under control and the joints are acted by the actuators and the center of mass of the manipulator-like spacecraft moves. Figure 11 enhances the control actions on the arms during the manipulator operation and the associated perturbation on the attitude motion. Every time an actuation occurs on a joint, the attitude is disturbed and corrected by attitude control.

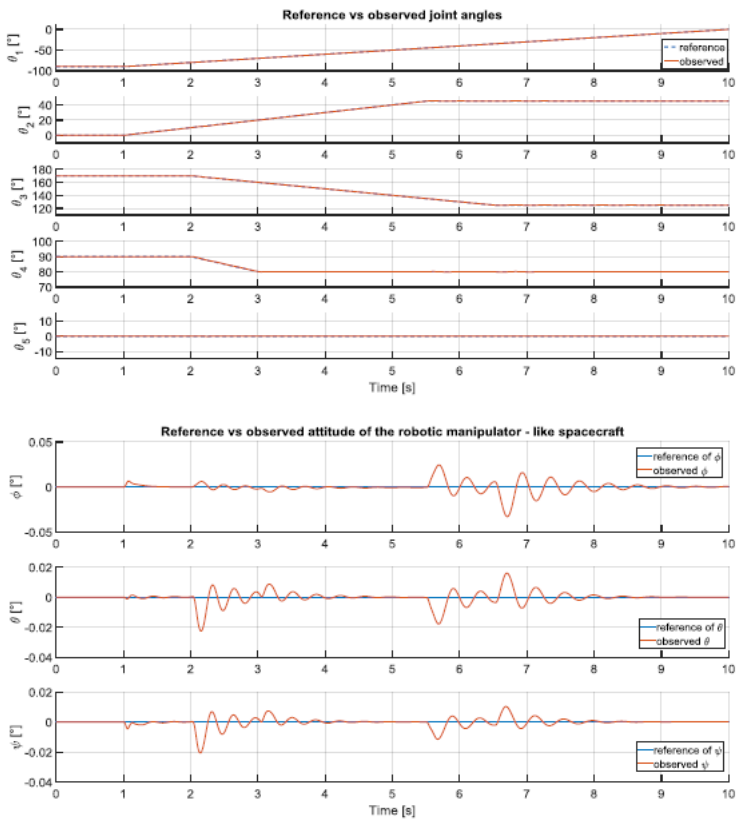


Fig. 11 Manipulator like-spacecraft is operating causing perturbation on the attitude that is corrected by the attitude control action.

Figure 12 enhances the control effort to move the arm. Such control effort yields important information for the specification of appropriate joint actuators.

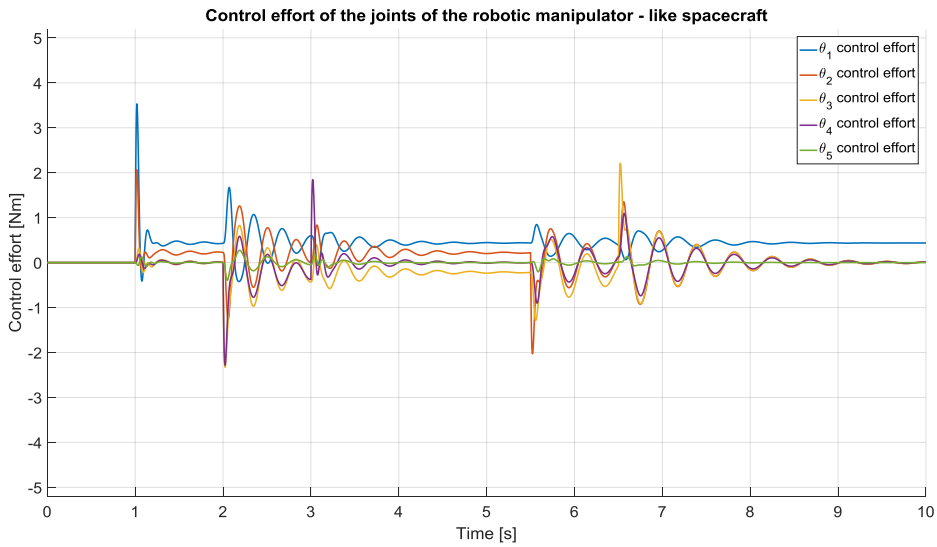


Fig. 12- Control effort to operate the manipulator

Figure 13 shows the control effort of the arm to correct the attitude while the arm is being acted by the joint actuators and the system center o f mass is moving.

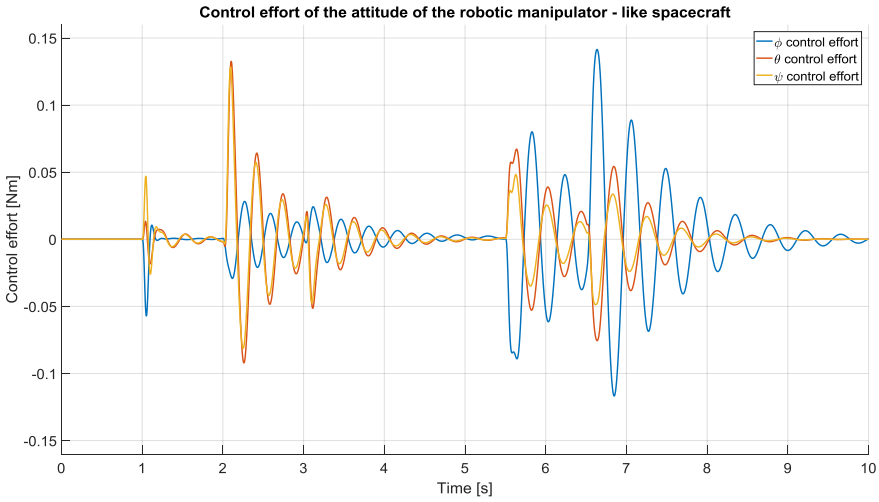


Fig. 13 Control effort to correct the attitude while the arm is moving

5. Conclusion

This paper presents a dynamic analysis of the motion of manipulator-like spacecraft in the microgravity environment while approaching to a target for RVD/B in the close approach range phase. The Lagrangian formulation with the Lagrange's multiplier and a Rayleigh dissipation function is used to generate a system of equations with extra coordinates so as to obtain information of the reaction torque/forces in the manipulator joints during the operation. The mathematical model considers also the instantaneous center of mass once it moves while the robot arm moves its links toward the target. As the mathematical model is very complex it was developed some simulations to test the correctness of that model. The results are analyzed w.r.t. the configuration in which the attitude control is disabled and the manipulator is moving. Then it is carried out computer simulations with active attitude control, the robot arm is moving with consequent change of the center of mass.

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